

SEYED MOHAMMAD SALEHI

Robotics Engineer | AI/ML Engineer/Researcher | Electrical/Electronics Engineer

London, UK • +44 7584 070723 • ss06053@surrey.ac.uk • madvid.work@gmail.com
linkedin.com/in/madvid • github.com/madvidd • madvidd.github.io/madvid

PROFESSIONAL PROFILE

Robotics engineer, AI/ML engineer/researcher and electrical/electronics engineer; M.Sc. graduate in Computer Vision, Robotics and Machine Learning from the University of Surrey. Combines experimental ML research with robotics, control, embedded firmware and electronics. Experience includes motion forecasting, autonomous systems, PID and safety logic, UAV flight control, STM32 development, circuit design, system testing and hardware-software integration.

CORE TECHNICAL SKILLS

AI/ML & research

Python, PyTorch, machine learning, deep learning, computer vision, motion forecasting, time-series modelling, Mamba/state-space models, attention, Argoverse 2, model training, benchmarking, experimental design, evaluation metrics and ablation studies

Robotics & control

Robotics, autonomous navigation and systems, UAVs, PID and control-system design, safety interlocks, hardware-software integration, system testing and validation, parameter tuning, flight-control programming, MATLAB/Simulink, signal processing

Embedded systems

C/C++, embedded C, firmware development, STM32 microcontrollers, STM32Cube, Keil MDK, microprocessors, Arduino, Atmel Studio AVR, Code Composer Studio, Visual Studio, Jupyter, Conda, Colab

Electrical & electronics

Electrical/electronic engineering, analogue/digital circuit design and fabrication, MOSFET/BJT switching, voltage/current regulation, embedded systems, IoT, FPGA/ASIC, VHDL, Vivado, Xilinx ISE, ADS, HSpice, PSpice, Proteus, Autodesk Fusion

ENGINEERING EXPERIENCE

Head Representative

Surrey Students' Union - Voice Zone, FEPS

Aug 2026 – Present

Guildford, UK

- Work in the Voice Zone as Head Representative for the Faculty of Engineering and Physical Sciences (FEPS).

Electrical Control Engineer

Rhizo PTX (freelance) / iTEK Project

Mar 2026 – Sep 2026

Surrey, UK

- Designed and integrated a hardware-software dashboard with PID control, parameter tuning, operating feedback, safety interlocks and operator controls.
- Engineered MOSFET- and NPN-transistor switching stages for voltage/current regulation and implemented the corresponding software interface.

Software Engineer - Embedded Systems

Sarmad Company (part-time)

Feb 2023 – Jul 2023

Tehran, Iran

- Developed embedded-system software in C using Visual Studio and Code Composer Studio.

Robotics and AI Trainer

Sigma Company (part-time)

Apr 2020 – Sep 2022

Tehran, Iran

- Taught robotics and AI, coached competition teams, and co-developed curricula, robot bodies, electronic circuits and starter board firmware.
- Helped organise robotics competitions at the University of Tehran (2021) and in India (2022).

Software Engineer - Embedded Systems

Cube Electronic (part-time)

Apr 2022 – Jun 2022

Tehran, Iran

- Developed C/C++ firmware for STM32 microcontrollers using STM32Cube and Keil MDK.

Telecommunications Analyst

XICOMM (part-time)

Jul 2021 – Nov 2021

Remote, USA

- Managed, tested and monitored Voice over IP calls as a remote Network Operations Centre team member.

EDUCATION

M.Sc. Computer Vision, Robotics and Machine Learning – Distinction

University of Surrey

Sep 2025 – Sep 2026

Guildford, UK

- Thesis: Uncertainty-Aware Attention, Geometry and State-Space Modelling for Streaming Motion Forecasting; evaluating DeMo, SEAM and SHARP on Argoverse 2 and extending SHARP with Mamba encoders.
- Final module average excluding thesis: **70.13%** (thesis mark pending).
- Module marks: Applied Machine Learning **85**; Robotics **78**; Research, Professionalism and Innovation **75**; Computer Vision and Pattern Recognition **70** (coursework **100/100**); Medical Robotics **67**; Advanced Topics in Computer Vision and Deep Learning **62**; Speech & Audio Processing & Recognition coursework **98/100**.

B.Sc. Electrical Engineering – Upper second class honours (2:1)

Shahid Beheshti University

Sep 2019 – Jul 2023

Tehran, Iran

- GPA: **77/100**; thesis: Simulation of a Self-Driving Car (92.5/100). Strong results in embedded systems, programming, microprocessors and linear control.

High School Diploma, Mathematics and Physics – A*

National Organization for Development of Exceptional Talents (Helli 4)

Oct 2016 – Jul 2019

Tehran, Iran

GPA: **96/100**.

SELECTED RESEARCH AND ENGINEERING PROJECTS

Human Motion Prediction for Safe Autonomous Navigation M.Sc. research, 2025 – 2026

- Benchmarking DeMo, SEAM and SHARP on the full Argoverse 2 motion-forecasting dataset using average displacement error, final displacement error and miss rate.
- Extending SHARP with Mamba state-space encoders and controlled attention ablations; assessing accuracy, efficiency, robustness and failure behaviour for autonomous-system deployment.

IMechE UAS Challenge 2026 - Team Peryton University of Surrey, 2025 – 2026

- Built and flew a UAV with a multidisciplinary team; contributed flight-control programming, parameter tuning, electronic-circuit design and fabrication; awarded a Certificate of Participation.

Battery SOC and SOH Estimation with Deep Learning Research project, 2024 – 2025

- Prepared lithium-ion battery measurements and evaluated SABRE- and RELIANT-based deep-learning architectures for state-of-charge and state-of-health prediction.

RESEARCH OUTPUT AND SELECTED CERTIFICATIONS

Research article: Enhancing SOC and SOH Estimation in Lithium-Ion Batteries through SABRE and RELIANT-Based Approach.

Certifications: Machine Learning Specialization (Stanford University/Coursera); Mathematics for Machine Learning and Data Science Specialization (DeepLearning.AI/Coursera); Self-Driving Car Engineer Training (Udacity); IMechE UAS Challenge 2026 Certificate of Participation.

ADDITIONAL TECHNICAL SKILLS

Robotics: ROS/ROS 2, SLAM, path planning, sensor fusion, Gazebo, OpenCV and Linux.

AI/ML and software: NumPy, pandas, scikit-learn, TensorFlow, JAX, feature engineering, hyperparameter tuning, Git, Docker, model deployment and MLOps.

Electrical/electronics: PCB design, Altium, KiCad, board bring-up, oscilloscopes, EMC/EMI and requirements verification.

Research: statistical analysis, reproducibility, experiment tracking, literature review and technical publication.

Creative: DJ and music production.

COMPLETE EXPERIENCE

Team Member (Rubix)

Surrey Students' Union, Guildford, UK

Sep 2026 – Present

Treasurer (society finances)

Surrey CompSoc

Mar 2026 – Present

Treasurer (society finances)

Surrey Walking Society

Mar 2026 – Present

Member (radio/3D projects)

Surrey EARS

Sep 2025 – Sep 2026

Member (innovation programme)

Surrey Enterprise Hub

Sep 2025 – Sep 2026

Volunteer (community support)

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Sep 2025 – Sep 2026

Teaching Assistant - Artificial Intelligence (AI support)

Prof. Nabavi, Shahid Beheshti University

Feb 2023 – Feb 2024

Teaching Assistant - Artificial Intelligence (AI support)

Prof. Aghaei, Shahid Beheshti University

Feb – Jul 2023

Editor-in-Chief (technical publishing)

Kahroba-ye-Beheshti, Shahid Beheshti University

Sep 2020 – Sep 2021

Member (UAS competition)

Team Peryton

Sep 2025 – Present

Treasurer (society finances)

University of Surrey Languages Society

Mar 2026 – Present

Course Representative (student liaison)

Surrey Students' Union

Sep 2025 – Sep 2026

Member (event support)

Surrey Stage Crew

Sep 2025 – Present

Member (space-engineering events)

Peryton Space

Sep 2025 – Sep 2026

Teaching Assistant - Microprocessors (lab support)

Shahid Beheshti University

Feb – Jul 2024

Teaching Assistant - Technical Language (EE English)

Shahid Beheshti University

Feb – Jul 2023

Research Secretary (research outreach)

EE Scientific Association, Shahid Beheshti University

Sep 2020 – Sep 2021

LANGUAGES AND TESTS

English: IELTS Academic 7.0 (2024: L7.5/W6.5/S6.5/R6.5; 2022: L7.5/W6.5/S6.5/R8.0); 12 years' formal study.

German: A1. **Persian:** Native. **GRE General:** taken 9 Jan 2024; Quantitative Reasoning: 155.

PROFESSIONAL REFERENCES

Professor Saber Fallah

1st M.Sc. Thesis Supervisor; Professor of Safe AI and Autonomy, University of Surrey.

s.fallah@surrey.ac.uk

Dr Gareth Beeston

iTEK Project Leader; Senior Project Officer, Surrey Student Enterprise, Innovation and Impact.

g.beeston@surrey.ac.uk

Ben Kyffin

Rhizo PTX Project Recommender; CTO and Co-founder, Rhizo PTX.

info@rhizoptx.com

Professor Kevin Wells

2nd Supervisor; Professor of AI in Human and Veterinary Healthcare, University of Surrey (CVSSP).

k.wells@surrey.ac.uk

Ash Stott

Rhizo PTX Project Recommender; CEO and Co-founder, Rhizo PTX.

info@rhizoptx.com